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# Department of Computer Engineering

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## Agentic Multi-LLM Validation System
## Label-Free Quality Assurance for Locally Hosted Large Language Models

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## Abstract

Large Language Models produce fluent text that is frequently wrong, and the usual remedy — asking a second model to grade the first — inherits well-documented pathologies of its own. This project asks a narrower, tractable question: **can the quality of a locally hosted question-answering system be validated when no human-annotated ground truth exists at all?**

The system answers each question twice using two independently prompted local models (LLaMA 3 and Mistral), then validates the pair rather than trusting either. An independent Verifier (phi3) and Judge (qwen3.5:9b) score both candidates **blind** on an eight-factor, 80-point rubric. Semantic similarity, cross-model agreement and external Wikipedia/Wikidata checks are blended into one 0–1 confidence score; anything below a calibrated threshold is regenerated automatically, and a sixth agent (Gemma 2 9B) synthesises the final answer. Everything runs offline, with no paid APIs.

Validation is **label-free by construction**. A self-generated 5,000-question corpus was produced and the decision threshold calibrated against censored data using a deliberately injected control group, ROC analysis with Youden's J, and a 2,000-sample bootstrap — giving an adopted operating point of $\tau = 0.72$, 95% CI [0.69, 0.75].

## Introduction

An LLM does not signal uncertainty the way a traditional component signals an error: a wrong answer is indistinguishable in tone from a right one. The dominant response is the **LLM-as-a-judge** pattern, which scales well but is a measurement instrument built from the same material as the thing it measures.

- **Human evaluation** — accurate, but expensive, slow and unscalable.
- **LLM-as-a-judge** — fast, but exhibits *position*, *verbosity* and *self-preference* bias large enough to reverse a verdict.

A second constraint shapes this work: much of the evaluation literature assumes human preference labels or a frontier commercial judge. Both fail for a department laboratory and for any deployment whose data cannot leave the institution. This project therefore restricts itself to **models running locally on a single workstation**, and builds validation that works without labels.

Crucially, the validation apparatus is then **turned on itself** and measured for the very biases the literature warns about. A system that cannot be audited this way is of limited scientific value.

## Literature Survey

**[1] MT-Bench & LLM-as-a-Judge** – Zheng et al. (NeurIPS 2023) established that model judges correlate with human preference in aggregate, and documented position and self-enhancement bias. Our work reproduces those biases locally and *quantifies* them on our own pipeline.

**[2] G-Eval** – Liu et al. (EMNLP 2023) used chain-of-thought form-filling for more stable numeric scores. Our eight-factor rubric with per-criterion justifications follows the same principle of structured scoring over free-form judgement.

**[3] Prometheus** – Kim et al. (ICLR 2024) showed an open, smaller model can approach proprietary grading behaviour, which is what makes a fully local judge credible.

**[4] BERTScore** – Zhang et al. (ICLR 2020). Embedding-based, but *reference-based*: unusable as a quality metric here. We use it only for cross-model agreement, never against a reference answer.

**[5] AutoGen** – Wu et al. (2023) popularised multi-agent role decomposition. We adopt the decomposition but reject reusing one model across roles, which silently destroys evaluator independence.

**[6] Youden's J** – Youden (1950), and **[7] the Bootstrap** – Efron & Tibshirani (1993): the statistical basis for choosing and bounding our decision threshold.

**Gap addressed:** a complete, reproducible account of building such a system under a hard no-ground-truth, no-external-API constraint, and then measuring its own bias with position, identity and length properly controlled.

## Objectives

- Build a **six-agent pipeline** in which generation and evaluation are performed by **different** models, running fully offline via Ollama with real-time SSE streaming.
- Define a **multi-signal confidence score** blending model-internal and external evidence so no single signal dominates.
- Gate an **automatic regeneration loop** on that score, with a bounded attempt count.
- **Calibrate the threshold from data** rather than intuition — accounting for the fact that the gate censors exactly the observations needed to calibrate it.
- Generate a corpus large enough for that calibration to be statistically meaningful, **without human annotation**.
- **Measure the validation apparatus itself** for position and self-preference bias, and correct what is found.
- Document measurement errors made along the way, including those caught before they reached a conclusion.

## Methodology

**Phase 1 – Multi-Agent Pipeline**

- FastAPI backend streaming every stage transition over Server-Sent Events.
- LLaMA 3 and Mistral generate independently via `asyncio.gather()` (langchain-ollama).
- Verifier (**phi3**, temp=0.1) and Judge (**qwen3.5:9b**, temp=0.1, reasoning disabled) each score *both* answers on the same **8-factor / 80-point** rubric.
- Combiner (**Gemma 2 9B**, temp=0.4) synthesises a ≤150-word answer — deliberately **not** one of the generators.

**Phase 2 – Bias Controls**

- **Brand-blind prompts:** evaluators never learn which system wrote which answer.
- **Position randomisation:** a fresh coin-flip per question assigns answers to slots; the true mapping is stored separately so per-model statistics stay recoverable.
- Rubric totals are **recomputed** from criterion scores — evaluators were observed stating totals that contradicted their own scores.

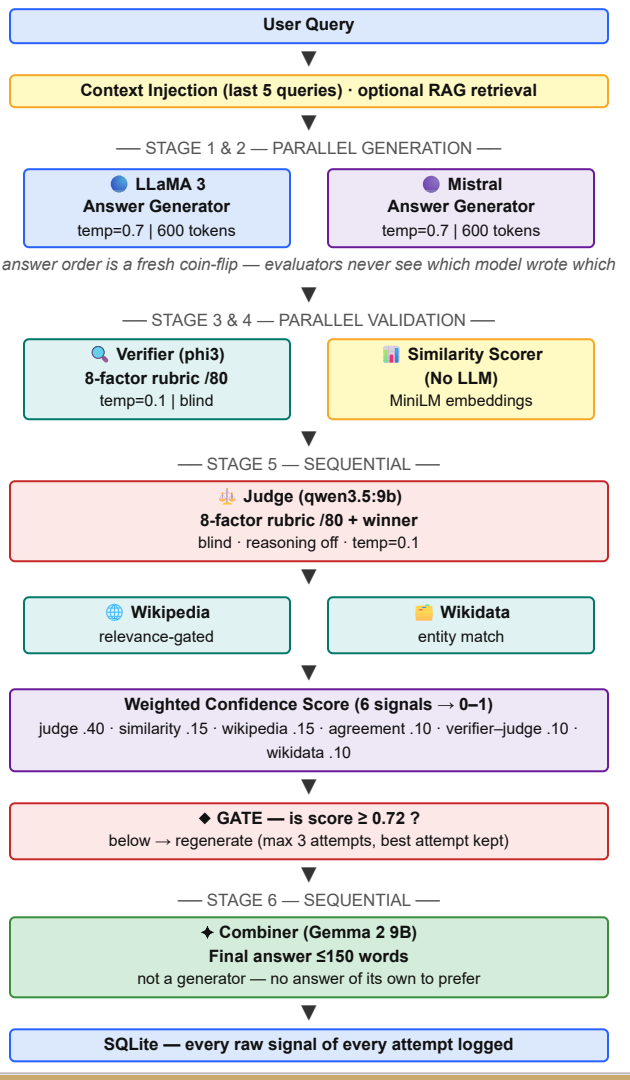
**Phase 3 – Label-Free Calibration**

- Answers above the gate are never regenerated, so the data are **censored**. A random fraction of *passing* answers is regenerated anyway, purely to measure — a **control group**.
- Three independent routes: two-proportion z-test, ROC with Youden's J, and a 2,000-sample bootstrap.

**Phase 4 – Auditing the Auditor**

- Embed both candidates and the final answer; measure which the final sits closer to.
- Cross **position × identity × judge verdict** to separate effects that are otherwise conflated; control for answer length.

## Flow Diagram




## Experimental Details

**Dataset: 5,000 questions** across 10 domains — Science & Technology, History & Geography, Health & Medicine, Economics & Finance, Philosophy & Ethics, Environment & Climate, Psychology & Behaviour, Society & Culture, Mathematics & Logic, Space & Astronomy. **8,432** attempt records; 5,000 final answers; mean 1.69 attempts per question.

**Models (all local via Ollama):** generators `llama3` and `mistral` (temp 0.7, 600 tok); Verifier `phi3:latest` (0.1, 700); Judge `qwen3.5:9b` (0.1, 900, reasoning off); Combiner `gemma2:9b` (0.4, 1000).

**Embedding Model:** all-MiniLM-L6-v2 (22.7M params, 384-dim, sentence-transformers), GPU-resident.

**Rubric (8 criteria × 10 = 80):** Factual Correctness, Question Relevance, Context Relevance, Faithfulness, Completeness, Hallucination-Free, Clarity, Consistency — applied identically by Verifier and Judge, which is what makes their agreement meaningful.

**Hardware:** single workstation, NVIDIA RTX 4090 (24 GB VRAM), Windows, Python 3.13. The six models exceed VRAM if co-resident, so the pipeline runs at concurrency 1 and the runtime evicts weights between stages.

**Statistical Methods:** two-proportion z-test; ROC with Youden's J; 2,000-sample bootstrap; Wilson intervals; inverse-probability weighting for a control rate that changed mid-run.

**Software Stack:** Python · FastAPI · Ollama · LangChain · sentence-transformers · scikit-learn · SciPy · SQLite · NumPy.

## Results

✅ **Threshold calibrated from data (n = 5,000):**

- Three independent routes agree: z-test **p = 0.011**; ROC point optimum **t = 0.70** (AUC **0.629**); bootstrap median **0.72**, 95% CI **[0.69, 0.75]**.
- Youden's J is near-flat across 0.70–0.74, so that band is statistically indistinguishable — the choice within it is policy, not statistics.
- **τ = 0.72 adopted** (bootstrap median): sensitivity **66.7%**, specificity **50.3%**. The earlier 0.75 sat at the very edge of the CI, trading specificity (30.2%) for recall.

🔍 **The validation apparatus audited itself:**

- The original Combiner was `llama3` — also a generator, so it was merging *its own writing* against Mistral's.
- **Identity effect: +18.1 pp** preference for its own answer, position held fixed ($p \approx 1.7\times10^{-89}$).
- **Position effect: +10.8 pp** toward whichever answer sat in prompt slot 1 ($p \approx 5.7\times10^{-53}$).
- **Length ruled out:** r = +0.029; the effect persists at **61.1%** among length-matched pairs.
- Mechanism is *selective obedience*: the Combiner followed the Judge **69.4%** of the time when the Judge picked its own answer, but only **49.3%** otherwise.
- Fixed by construction — Gemma 2 9B has no answer of its own among the two it merges.

🧪 **Evaluator discrimination verified:** the Verifier scores intact answers **77–80/80** and deliberately corrupted answers **0–8/80** — a **74.5-point** separation, confirming the evaluators discriminate rather than rubber-stamp.

> ⚠️ **Four measurement errors were caught before they reached a conclusion** — a model reporting a total contradicting its own scores; a "second opinion" signal secretly derived from the Judge; a sampling weight applied after the rate had changed mid-run; and restart-orphaned duplicate final rows. Each was found by checking a result that looked *too clean*.

## Conclusion

A locally hosted question-answering system **can** be validated without ground truth — with honest limits. Internal consistency, evaluator discrimination, relative improvement and threshold calibration are all achievable label-free. **Absolute factual accuracy is not**, and no such claim is made.

Key contributions:

- **A label-free validation methodology** using a deliberately injected control group to defeat the censoring that the regeneration gate itself creates.
- **A threshold derived three independent ways** that converge on one narrow band — τ = 0.72, CI [0.69, 0.75] — rather than chosen by intuition.
- **A measured decomposition of evaluator bias** into identity (+18.1 pp) and position (+10.8 pp) components with length controlled — which *overturned this project's own prior design decision.*
- **A fully offline six-agent system** in which no model both writes an answer and evaluates or merges it.

The system is best understood not as a correctness oracle but as a **calibrated quality gate**: it reliably separates weaker answers from stronger ones and acts on that separation, while remaining honest about what it cannot certify. **Open problem:** the +10.8 pp position effect is mitigated but not eliminated; extractive selection (MMR) would remove it by construction.

## References

[1] L. Zheng et al., "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," *NeurIPS*, 2023.

[2] Y. Liu et al., "G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment," *EMNLP*, 2023.

[3] S. Kim et al., "Prometheus: Inducing Fine-grained Evaluation Capability in Language Models," *ICLR*, 2024.

[4] T. Zhang et al., "BERTScore: Evaluating Text Generation with BERT," *ICLR*, 2020.

[5] Q. Wu et al., "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," *arXiv:2308.08155*, 2023.

[6] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *NeurIPS*, 2020.

[7] N. Reimers & I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," *EMNLP*, 2019.

[8] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950.

[9] B. Efron & R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

[10] J. Carbonell & J. Goldstein, "The use of MMR, diversity-based reranking...," *ACM SIGIR*, 1998.

[11] A. Q. Jiang et al., "Mistral 7B," *arXiv:2310.06825*, 2023.

[12] Gemma Team, "Gemma 2: Improving Open Language Models at a Practical Size," *arXiv:2408.00118*, 2024.